

Natural Language Processing

Lab 4: Standard document classification with scikit-learn

2024/25

1. For this week's lab session we will follow the bag-of-words text classification tutorial provided by the popular python machine learning library scikit-learn.

(a) Follow the tutorial here:

https://scikit-learn.org/stable/auto_examples/text/plot_document_classification_20newsgroups.html

(b) The tutorial provides Jupyter notebook and source code files. Download whichever you prefer, and step through the code.

(c) For each section of the tutorial, refer to the documentation to understand the functions and parameters being used. This link brings you to the TfidfVectorizer documentation:

https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html

(d) Find the Naive Bayes classifier used in the tutorial. Look up its documentation, and consider how it compares to the theory and example we covered in class.